

STAT 301 In-class Activity No. 4

Consider the “wage” dataset in Activity No. 3 again. We studied the relationships between WAGE and EDUCATION, SEX, and OCCUPATION.

- (a) Compared with the results in activities No. 3, does the relationship between WAGE and EDUCATION depend on SEX? In either case (i.e., yes or no), are the interpretations of parameter β_1 (the parameter associated with EDUCATION) the same or not?
- (b) Does the relationship between WAGE and EDUCATION depend on OCCUPATION? Why?
- (c) Fit an additive model with WAGE as the response and EDUCATION, SEX, and OCCUPATION as predictors. Perform model diagnostics based on residual plots and QQ plot. Does the model fit the data well? Any potential issues?